

# SMART INDIA HACKATHON 2026



## TITLE PAGE

**Problem Statement ID** – SIH26028

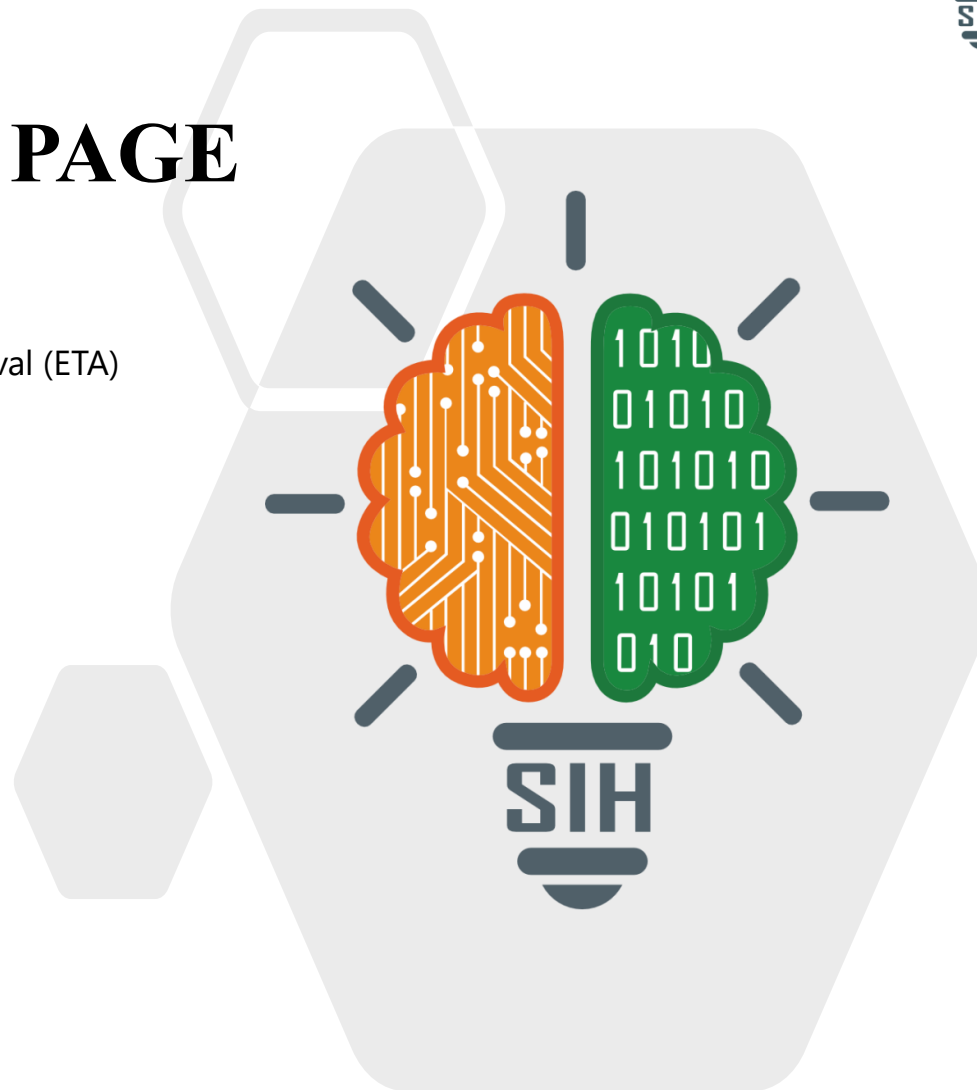
**Problem Statement Title** – Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains

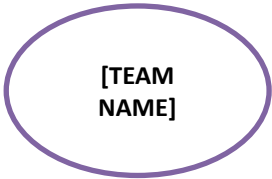
**Theme** – Smart Automation

**PS Category** – Software

**Team ID** – [TEAM ID]

**Team Name (Registered on portal)** – [TEAM NAME]





[TEAM  
NAME]

# RAILCAST — an arrival window, not a single number



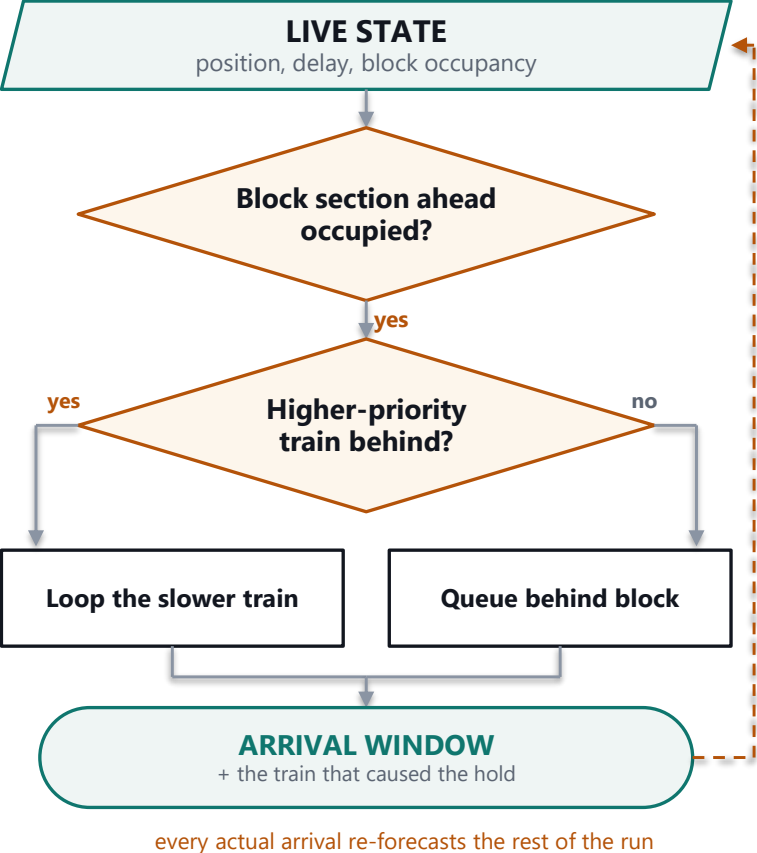
## THE PROBLEM

- Today's ETA repeats the delay a train already has and trusts timetable padding.
- It cannot see a conflict building ahead, so a held train surprises everyone behind it.
- One number, no confidence — nobody knows when it is safe to act.

## HOW IT WORKS — ONE FORECAST

- A Free-run time**  
physics of each section
- B Conflict simulation**  
advance every train on the block graph
- C Learned residual**  
corrects what the rules cannot
- D Calibrated window**  
audited 80% coverage

## THE DECISION THE SIMULATOR RESOLVES

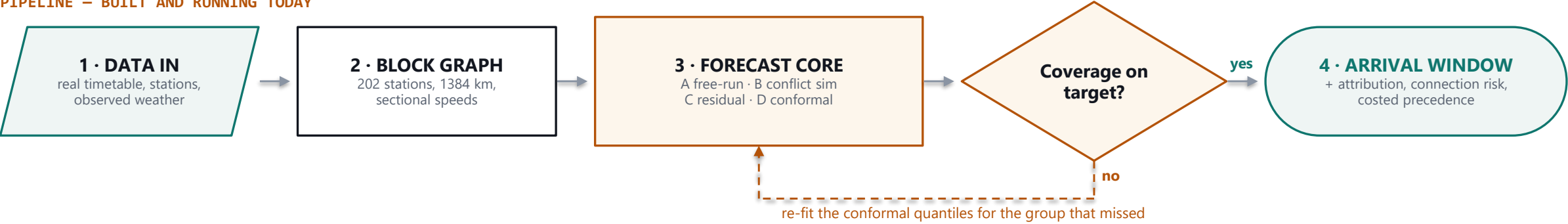


## WHAT MAKES IT DIFFERENT

- Forecasts the cause, not the clock**  
Trains advanced over the real block-section graph until they conflict — so the ETA moves for a reason we can name.
- States how sure it is, and audits it**  
80% window, coverage checked per segment, horizon and time of day — not pooled once and asserted.
- Names the blocking train**  
Attribution separates a controller's loop from a physical queue behind an occupied block.
- Checks itself against reality**  
A live feed collects real arrivals; the model is re-fitted to them and reports where it still misses.

Working prototype — every stage above runs today.

PIPELINE – BUILT AND RUNNING TODAY



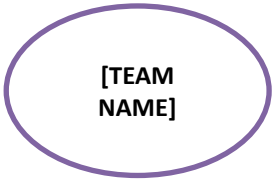
BUILT AND MEASURED – THIS REPOSITORY

|                                                                                           |
|-------------------------------------------------------------------------------------------|
| <b>Language and models</b>                                                                |
| Python 3 · LightGBM (free-run, quantile residual) · scikit-learn · NumPy / pandas         |
| <b>Simulation</b>                                                                         |
| Custom event-driven block-graph simulator (heapq priority queue) – no graph library       |
| <b>Uncertainty</b>                                                                        |
| Own conformalised quantile regression, Mondrian groups by horizon × segment × time of day |
| <b>Data adapters</b>                                                                      |
| Indian Railways working timetable (CC0) · Open-Meteo ERA5 weather · RailRadar live feed   |
| <b>Attribution</b>                                                                        |
| LightGBM split gain, plus the blocking train and cause recorded by the simulator          |

PLANNED FOR DEPLOYMENT – NOT BUILT YET

|                                                                                       |
|---------------------------------------------------------------------------------------|
| <b>Serving</b>                                                                        |
| FastAPI · Redis Streams · PostgreSQL + PostGIS · Docker / Kubernetes, sharded by zone |
| <b>Passenger interface</b>                                                            |
| React · MapLibre GL for app and station displays                                      |
| <b>Reaching every passenger</b>                                                       |
| SMS and IVR (139) in regional languages, for travellers without a smartphone          |
| <b>Between-station sensing</b>                                                        |
| Opt-in passenger-phone motion to close the 40-60 minute reporting gap                 |
| <b>Operations</b>                                                                     |
| Drift monitor, scheduled retraining, and a shadow-mode pilot beside the existing ETA  |

The prototype runs end to end on one commodity laptop with no GPU: 0.46 ms per station forecast, 0.32 s to re-forecast all 688 station arrivals on the corridor.



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# FEASIBILITY AND VIABILITY



## WHY IT IS FEASIBLE

### It already runs

The whole pipeline executes end to end on a laptop — corridor, simulator, models, calibration, evaluation.

### No new trackside hardware

It consumes position data Indian Railways already produces, through a swappable adapter.

### Cheap to serve

Boosted trees, no GPU. 0.46 ms per station forecast on one commodity node.

### Free, open inputs

Timetable and station master are CC0; ERA5 weather needs no key.

### Degrades safely

With the learned layer off, the rule-based simulator still produces a forecast.

## RISKS AND HOW WE RETIRE THEM

### The delay model is not yet fitted to reality

Validated against 499 real arrivals: our tail is too heavy. The collector runs daily; refit as data accumulates.

### No public archive of past arrivals

None exists, so history is accumulated by polling a live feed — already collecting.

### Error compounding over a 40-hour run

Retired: each station is forecast in one shot from current state, never chained.

### Coverage good on average, bad in a season

Retired: calibration is conditioned on segment, horizon and time of day, and audited per cell.

### Controllers distrust a black box

Every forecast names the blocking train and section. Advisory only — the system issues no movement order.

## STATUS TODAY

✓ Corridor from the real timetable

✓ Block-graph conflict simulator

✓ Four-stage forecast core

✓ Calibrated 80% window, audited

✓ Live feed connected, collecting

✓ Validation against real arrivals

○ Delay model fitted to real data

○ Serving stack and passenger apps

○ Shadow-mode pilot on a corridor

Six of nine stages are built and measured. The remaining three are engineering, not research.

## ROADMAP



**60%**  
**lower mean ETA error**  
26.9 → 10.9 min

**62%**  
**lower P90 error**  
77 → 29 min

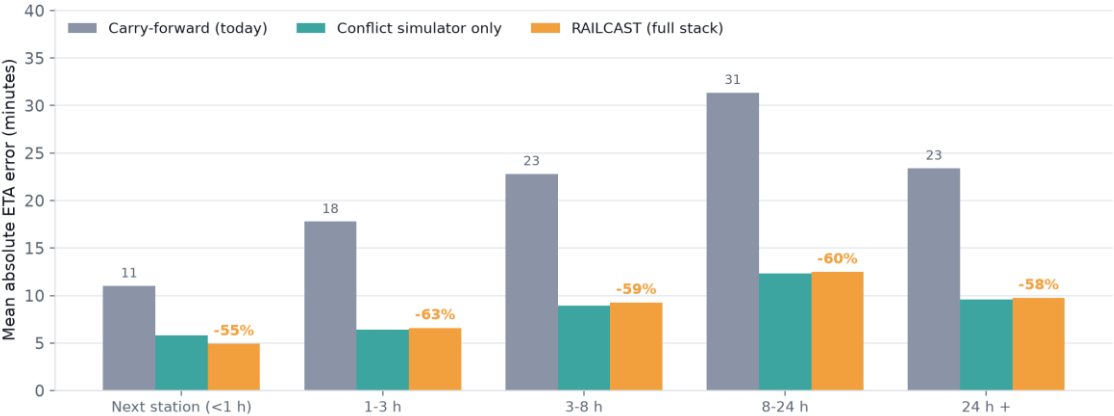
**78.4%**  
**coverage achieved**  
of the nominal 80% window

**40 h**  
**forecast horizon**  
station by station

Measured on 73,268 forecasts from held-out days, against carry-forward computed on exactly the same rows.

**Error by forecast horizon, not one pooled number**

Carry-forward is adequate for the next station. It collapses beyond a few hours - which is the horizon passengers actually plan against.



Real Indian Railways working timetable and stations (datameet, CCO) with observed ERA5 weather - delay generation is still modelled, not measured

Carry-forward is adequate for the next station and collapses beyond a few hours — the horizon passengers actually plan against.

**WHO FEELS THE DIFFERENCE**

**Passengers**

An arrival window to plan around, the odds of making a connection, and when to leave home.

**Section controllers**

Conflicts surfaced before they happen, with the blocking train named and the cost of each precedence call.

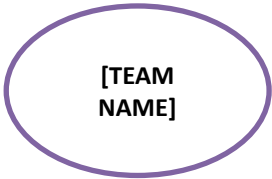
**Station operations**

Platforms, cleaning turnaround and crew duty planned against a forecast with a stated confidence.

**Feeder and logistics**

Bus, taxi and parcel operators schedule to the real arrival instead of idling at the kerb.

**Honest scope.** These are prototype results on a simulated corridor built from the real timetable. Delay generation is modelled, and validation against 499 real arrivals shows it is not yet right — which is why the calibration loop exists.



[TEAM  
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# RESEARCH AND REFERENCES



## DATA ACTUALLY USED

### Indian Railways working timetable

datameet/railways, CC0 — 8,697 stations, 5,208 trains, 417,080 scheduled calls

### Weather

ERA5 reanalysis via Open-Meteo — observed daily, per corridor segment, no API key

### Live running position

RailRadar REST API — 499 observed arrivals collected across 10 trains so far

### Route geometry

Station coordinates scaled to the published route distance; 1384 km reproduced exactly

## SCALE OF DEPLOYMENT

### Prototype

New Delhi – Mumbai Central, 1384 km, 202 stations, 44 trains — built and evaluated

### Zonal pilot

One zone, one control office, one model shard

### National

Order of 13,000 coaching trains a day (Indian Railways published figure — confirm before submission)

## RESEARCH WE BUILD ON

1. R. Oneto et al., Train delay prediction systems: a big data analytics perspective, Big Data Research 11, 2018.
2. P. Kecman, R. M. P. Goverde, Predictive modelling of running and dwell times in railway traffic, Public Transport 7, 2015.
3. F. Corman, P. Kecman, Stochastic prediction of train delays in real time using Bayesian networks, Transp. Research Part C 95, 2018.
4. Y. Romano, E. Patterson, E. Candès, Conformalized quantile regression, NeurIPS 32, 2019.
5. V. Vovk, A. Gammerman, G. Shafer, Algorithmic Learning in a Random World, Springer 2005 — Mondrian conditional validity.
6. G. Ke et al., LightGBM: a highly efficient gradient boosting decision tree, NeurIPS 30, 2017.
7. The Digital Personal Data Protection Act, 2023 (India).

## WHERE WE ADD TO THIS

None of the above combines a conflict-resolution simulator over the real block-section graph, a direct multi-horizon learned residual and conditionally calibrated intervals — then validates the result against live Indian Railways running data.

**Project links** code and reproducible results · simulation videos · evaluation tables and charts — [REPOSITORY LINK]